

List of publication and products

Prof. Dr. Pierros Ntelis

May 16, 2026

Via, a 14 *h*-index, a 16 *i10*-index, and total citations: 4002 ([Google Scholar 2026](#)), my most important work is represented in the following papers:

Accepted

- [1] P. Yovkochev and **Ntelis, P.**, 2026. Evolution and Mass Decay of a Gaussian Pulse in Time-Fractional Diffusion: An L1-Implicit Scheme Analysis. *Turanian Journal*, vol. 2, no. 1, pp. 020103-1–020103-7. [URL](#), Contribution: Supervision, and co-design of analysis. (Citations: 0)
- [2] **Ntelis, P.** 2025. Advancing Tensor Theories. Symmetry: Mathematics: Advances in Topology and Algebraic Geometry. [URL](#) (Citations: 2) ★
- [3] **Ntelis, P.** 2025. Advanced manifold-metric pairs. Mathematics. [ArXiv](#), [URL](#) (Citations: 1) ★
- [4] **Ntelis, P.** & J. L. Said, 2025. Analytical *poly* Λ CDM dynamics. PDU.[URL](#) (Citations: 1) ★
- [5] E.D. Valentino, J. L. Said, ..., **Ntelis, P.** ..., et al. 2025. The CosmoVerse White Paper: Addressing observational tensions in cosmology with systematics and fundamental physics. Contribution: Writing on extended theories of gravity, scientific review, and suggestions for further development. PDU [URL](#) (Citations: 472)
- [6] Ntelis, P. 2022. A (D_τ, D_x) -manifold with N -correlators of N_t -objects. IJGMMP, arXiv e-prints arXiv:2209.07472 [URL](#), (Citations: 0)
- [7] **Ntelis, P.** & J. L. Said 2025. Exploring ϕ CDM model dynamics. EPJC, arXiv:2502.03486v1 [URL](#), (Citations: 4)
- [8] **Ntelis, P.** & J. L. Said 2025. Simple $\phi\Lambda$ CDM dynamics. IJGMMP, arXiv:2502.03486v2 [URL](#), (Citations: 3)
- [9] S.Casas, ..., **Ntelis, P.**, and 128 colleagues 2025. Euclid: Constraints on $f(R)$ cosmologies from the spectroscopic and photometric primary probes. A&A, [URL](#) (Citations: 27) Contribution: Scientific review, software development, and contribution on discussions for further research.
- [10] Euclid Collaboration, and 1137 colleagues 2024. Euclid. I. Overview of the Euclid mission. A&A, arXiv:2405.13491 [URL](#), (Citations: 785) Contribution: Scientific review and contribution on discussions for further research.
- [11] **Ntelis, P.** & Morris, A. 2023. Functors of actions. Foundations of Physics, arXiv:2010.06707 [URL](#), (Citations: 24) ★
- [12] Göksel Karavaşlı, N., ... **Ntelis, P.**, and 10 colleagues 2022. Optimal 1D Ly α Forest Power Spectrum Estimation – II. KODIAQ, SQUAD & XQ-100. MNRAS [URL](#), (Citations: 58), Contribution: Scientific review and contribution on discussions for further research and software development.
- [13] Hamaus, N., ... **Ntelis, P.**, and 128 colleagues 2022. Euclid: Forecasts from redshift-space distortions and the Alcock-Paczynski test with cosmic voids. A&A arXiv e-prints [URL](#), (Citations: 95), Contribution: Scientific review and contribution on discussions for further research and software development.
- [14] Ilić, S., ... **Ntelis, P.**, and 207 colleagues 2021. *Euclid* preparation: XV. Forecasting cosmological constraints for the *Euclid* and CMB joint analysis. A&A arXiv e-prints [URL](#), (Citations: 104), Contribution: Scientific review and contribution on discussions for further research. ★
- [15] Alam, S., ... **Ntelis, P.**, and 33 colleagues 2020. Towards testing the theory of gravity with DESI: summary statistics, model predictions and future simulation requirements. JCAP [URL](#),

- (Citations: 75), Contribution: Scientific review and contribution on discussions for further research and software development. ★
- [16] Tutusaus, I., ... **Ntelis, P.**, and 89 colleagues 2020. Euclid: The importance of galaxy clustering and weak lensing cross-correlations within the photometric Euclid survey. *Astronomy and Astrophysics* 643, [URL](#), (Citations: 92), Contribution: Scientific review and contribution on discussions for further research.
- [17] **Ntelis, P.**, A. Ealet, S. Escoffier, J.-C. Hamilton, A. J. Hawken, J.-M. Le Goff, J. Rich, and A. Tilquin 2018. The scale of cosmic homogeneity as a standard ruler. *Journal of Cosmology and Astroparticle Physics* 2018, [URL](#), (Citations: 28)
- [18] Blanton, M. R., ... **Ntelis, P.**, and 361 colleagues 2017. Sloan Digital Sky Survey IV: Mapping the Milky Way, Nearby Galaxies, and the Distant Universe. *The Astronomical Journal* 154, [URL](#), (Citations: 2122), Contribution: Scientific review and contribution on discussions for further research. ★
- [19] **Ntelis, P.**, and 22 colleagues 2017. Exploring cosmic homogeneity with the BOSS DR12 galaxy sample. *Journal of Cosmology and Astroparticle Physics* 2017, [URL](#), (Citations: 113) ★
- [20] Laurent, P., ... **Ntelis, P.**, and 17 colleagues 2016. A $14 \text{ h}^{-3} \text{ Gpc}^3$ study of cosmic homogeneity using BOSS DR12 quasar sample. *Journal of Cosmology and Astroparticle Physics* 2016, [URL](#), (Citations: 99), Contribution: Scientific review and contribution on discussions for further research and software development.

Under review

- [1] **Ntelis, P.**, et al. 2019. Cosmological constraints from cosmic homogeneity. (Cites: 4) [URL](#)
- [2] **Ntelis, P.** 2025. A probabilistic and expanding Universe. [URL](#), (Cites: 2)
- [3] **Ntelis, P.** 2023. Advancing category and tensor theories. [URL](#), (Cites: 2)
- [4] **Ntelis, P.** 2025. CosmoFATs. [URL](#), (Cites: 2)
- [5] **Ntelis, P.** 2025. CosmoFATs Dynamics. [URL](#) (Cites: 0)
- [6] **Ntelis, P.** 2025. A cosmic Schiff's conjecture test. [URL](#), , [DOI](#), (Cites: 0)
- [7] **Ntelis, P.** 2025. A universal foundational theory. [URL](#), (Cites: 0)
- [8] **Ntelis, P.** 2026. · A dot. [URL](#), [DOI](#), (Cites: 0)
- [9] **Ntelis, P.** 2026. Analytical sub-poly Λ CDM through cosmic obs. [URL](#), [DOI](#), (Cites: 0)
- [11] **Ntelis, P.** 2026. A foundational analysis of local functional calculus [URL](#), [DOI](#), (Cites: 0)
- [11] **Ntelis, P.**, B.J. Ahmedov, C.Yuan, 2026. A cosmic femtolensing Hawking radiation signature [URL](#), [DOI](#), (Cites: 0)

Outreach	2022- n_{ow} , Universions Youtube Channel
	2025 , Utube , Comet 3i ATLAS: Interstellar Halloween Secrets! (6')
	2025 , Utube , Are there Aliens ? Drake's Equation ! (7')
	2025 , Utube , Quantum Tunnelling in Your Palm! (1')
	2025 , Utube , Advanced Manifold-Metric Pairs (10')
	2025 , Utube , A sunny watery magic (33'')
	2025 , Utube , Advanced Euler's identity (3')
	2025 , Utube , Analytical <i>poly</i> Λ CDM dynamics 101 (5' talk)
	2025 , Utube , Advancing Tensor Theories 101 (10' talk)
	2025 , Utube , Quintessence 101 (5' talk)
	2024 , Utube , Introduction cosmology for everyone (1h talk)
	2023('24), IBS of Provence , Neuroscience and cosmology for high school students (3h)
	2023 , IBS of Provence , Math and physics from CPT for high school students (3h)
	2022 , IBS of Provence , Mathematics and physics for high school students (3h)

Conferences	2026, HIT, SoP, Advanced math in cosmology (1h Seminar)
Talks/Posters	2026, HIT, SoP, A note on Baye's data analysis (1h Seminar)
Seminars	2025, The Dark Side of the Universe, FAT dynamics to cosmology (25' Talk)
	2025, Cosmo From Home, Advanced math in cosmology (Talk)
	2024, 50 yr of Horndeski at PI, A probabilistic expanding universe under FA , (Poster)
	2024, Cosmo From Home, A probabilistic expanding universe under FA , (25 min talk)
	2023, NEB, New avenues of functors of actions theories , (Talk 20')
	2023, Cosmo From Home, New avenues on Functors of Actions , (25 min talk)
	2023, NTUA HET, New theories in cosmology and particle physics , (1h Seminar)
	2022, CERN TH DPT Visit, Functors of Actions (5' Blackboard Talk)
	2021, Euclid Science Talk, Overview of impact of interlopers on PNG (1h Talk)
	2020, Atelier D.E., PNG with interlopers in Euclid , (20' Talk) HVT
	2020, Atelier D.E., Perturbation Theory of LSS, code: CLASS-PT , (20' Talk)
	2020, SEMFE, NTUA, LSS: Theory and Observations , (40' Seminar)
	2019, 30th SYMPOSIUM, Fractality from Multiple Tracers , (15' Talk)
	2019, Euclid IAS Paris, Systematics on PNG , (20' Talk)
	2019, Euclid Consortium Helsinki, Primordial Non-Gaussianity , (15' Talk)
	2019, eBOSS/DESI France, Cosmology with Cosmic Homogeneity (15' Talk)
	2019, Aix-University, Homogeneity with Multitracers , (2h Seminar)
	2019, Aix-University, A note on lss, Summer School , (6h Seminar)
	2018, Euclid GC-WL, Ameliorating cosmology with homogeneity , (15' Talk)
	2018, Colloquium of D.E., Cosmic Homogeneity as a standard ruler , (20' Talk)
	2018, R. Moriond, Euclid:Homogeneity in search of the Dark Sector , (Poster) HIC
	2018, UiO, What can fractality tell us about the universe? , (1h seminar)
	2017, P.O.N.T. in Cosmology, Is the universe homogeneous? , (20' Talk)
	2016, R. Moriond, The homogeneity scale of the universe , (15' Talk)
	2015, UnivEarthS School, Santorini, LSS and homogeneity , (Poster)
	2015, Rencontres de Jeunes Physicienne, Homogeneity (Video) , (15' Talk)
	2014, Elbereth Conference, IAP, The homogeneity of the universe (15' Talk)
	2014, UnivEarthS School, The homogeneity of the universe , (3' Talk)

Software

- [1] P.Ntelis (2017-today): [Github contributions](#)
- [2] P.Ntelis (2017-today): [COSMOPIT](#)
- [3] P.Ntelis (2018-2021): [Gitlab private Euclid softwares](#)
- [4] P.Ntelis (2018): [MCMC-posterior-example](#)
- [5] P.Ntelis (2018): [CoHo](#)
- [6] P.Ntelis (2018): [ANALYSE-SN-magnitude-redshift-curve](#)
- [7] P.Ntelis (2018): [Extract-Cosmology-with-the-BAO-peak-position](#)
- [8] P.Ntelis (2020): [AofEFT](#)
- [9] P.Ntelis (2023): [Tutorial on particle/galaxy simulations and coordinate transformation](#)
- [10] P.Ntelis (2025a): [Standard \$\phi\$ CDM dynamics](#)
- [11] P.Ntelis (2025b): [Simple quintessence dynamics](#)
- [12] P.Ntelis (2025c): [Analytical \$poly\Lambda\$ CDM dynamics](#)